



ALL I WANT FOR CHRISTMAS IS... ...AN ACTUARY

Health Claims Forum
London, 8 December 2011

Presented by: [Jonathan Hughes](#)

“Actuary draws tenuous link” (In other news: “Pope Catholic”)

What Simon asked
me to do today

A somewhat
tortuous, tenuous
trajectory

My presentation

“Do something Christmassy”

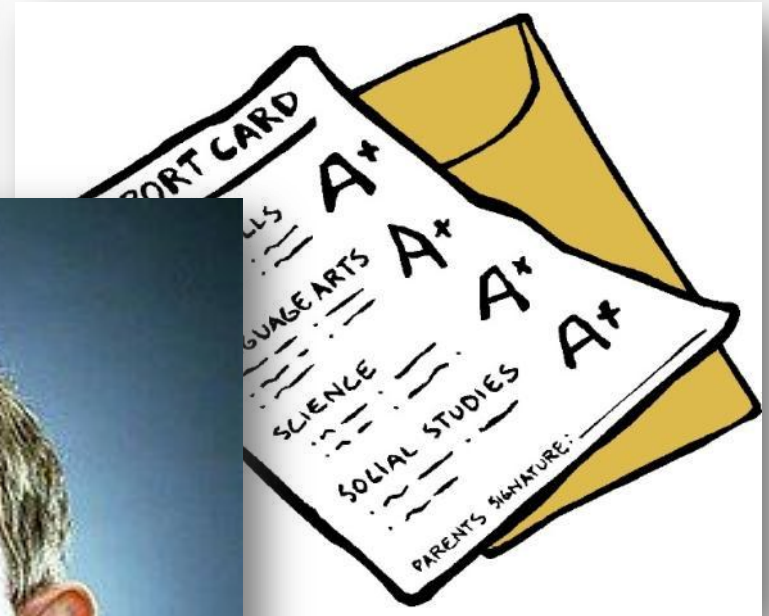


Havinden

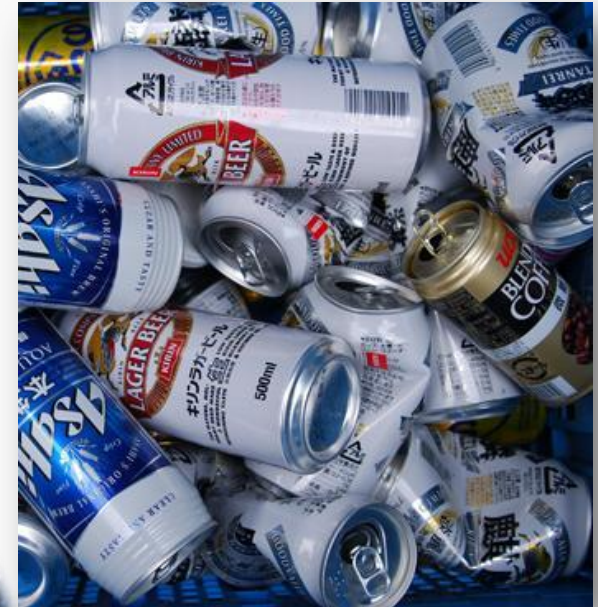


Claus

Santa Claus assessing Christmas claims



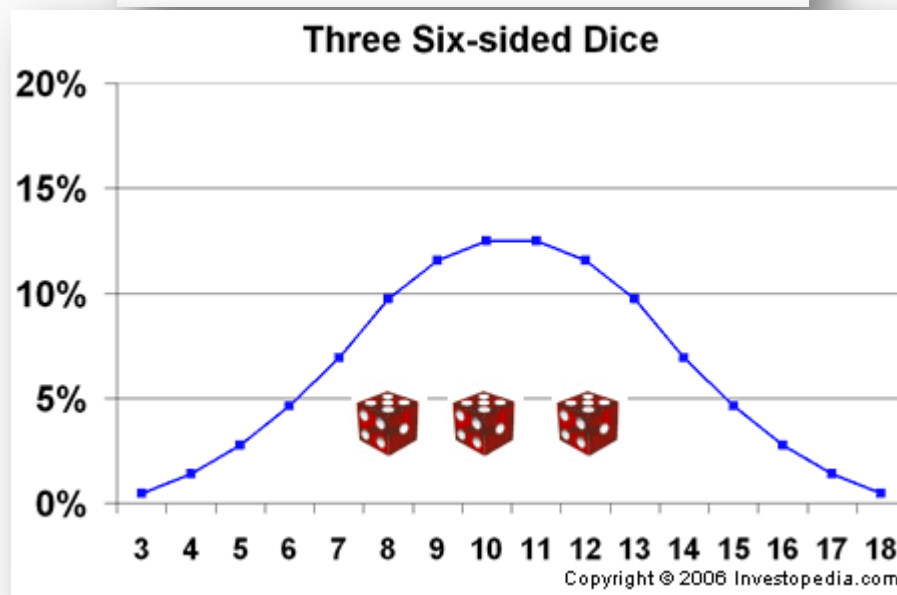
A man's approach to Christmas shopping



How lucky Simon is



$$P(a) = \frac{\sum_{i=0}^{a-1} \left(\frac{1-p}{p} \right)^i}{\sum_{i=0}^{a+b-1} \left(\frac{1-p}{p} \right)^i}$$



What I'm telling you

We're going to play a game

A volunteer will pick a present
from Simon's sack

I'll show you how I'd work out
the intended recipient

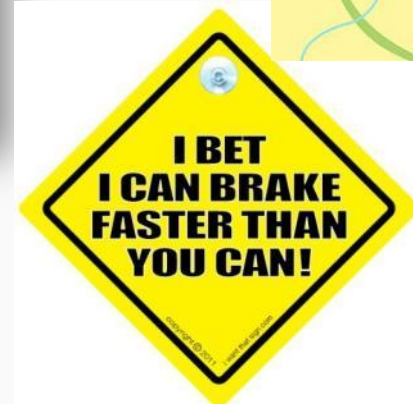
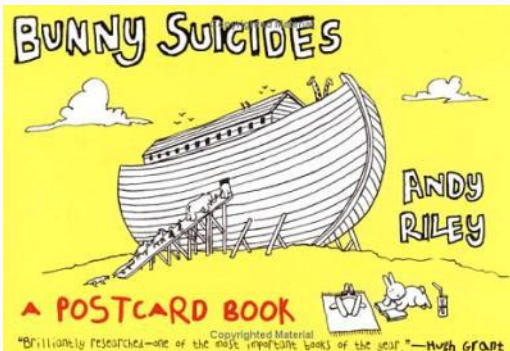
What you'll tell your boss

An actuary presented at the
HCF

He showed us various
different data sources

Some of these could be useful
in my day-to-day job

Simon's sack





Uncle Ian

Lives in some godforsaken
part of East London (Barking)

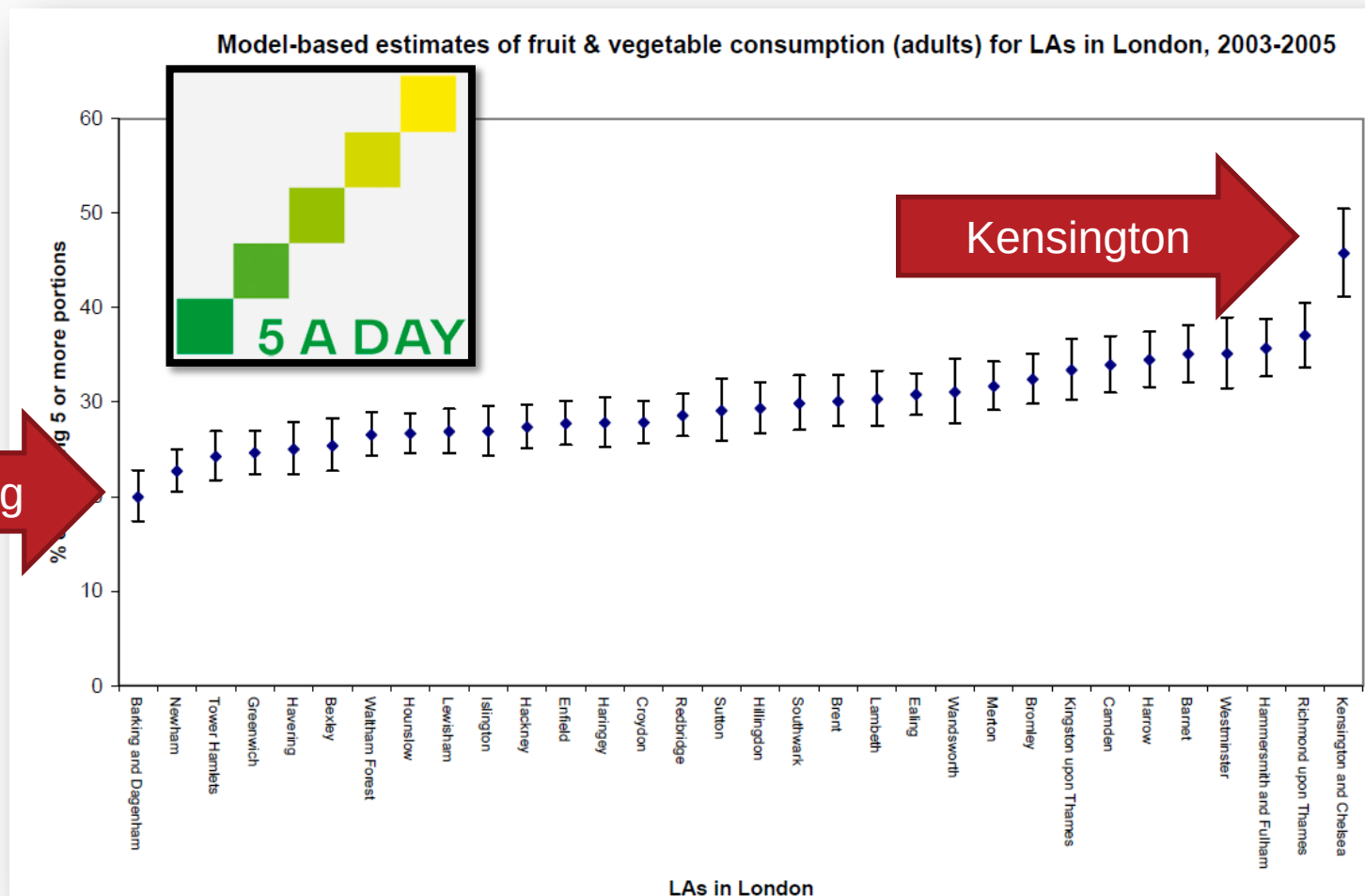


Uncle Mike

Lives in a nobby part of West
London (Kensington)

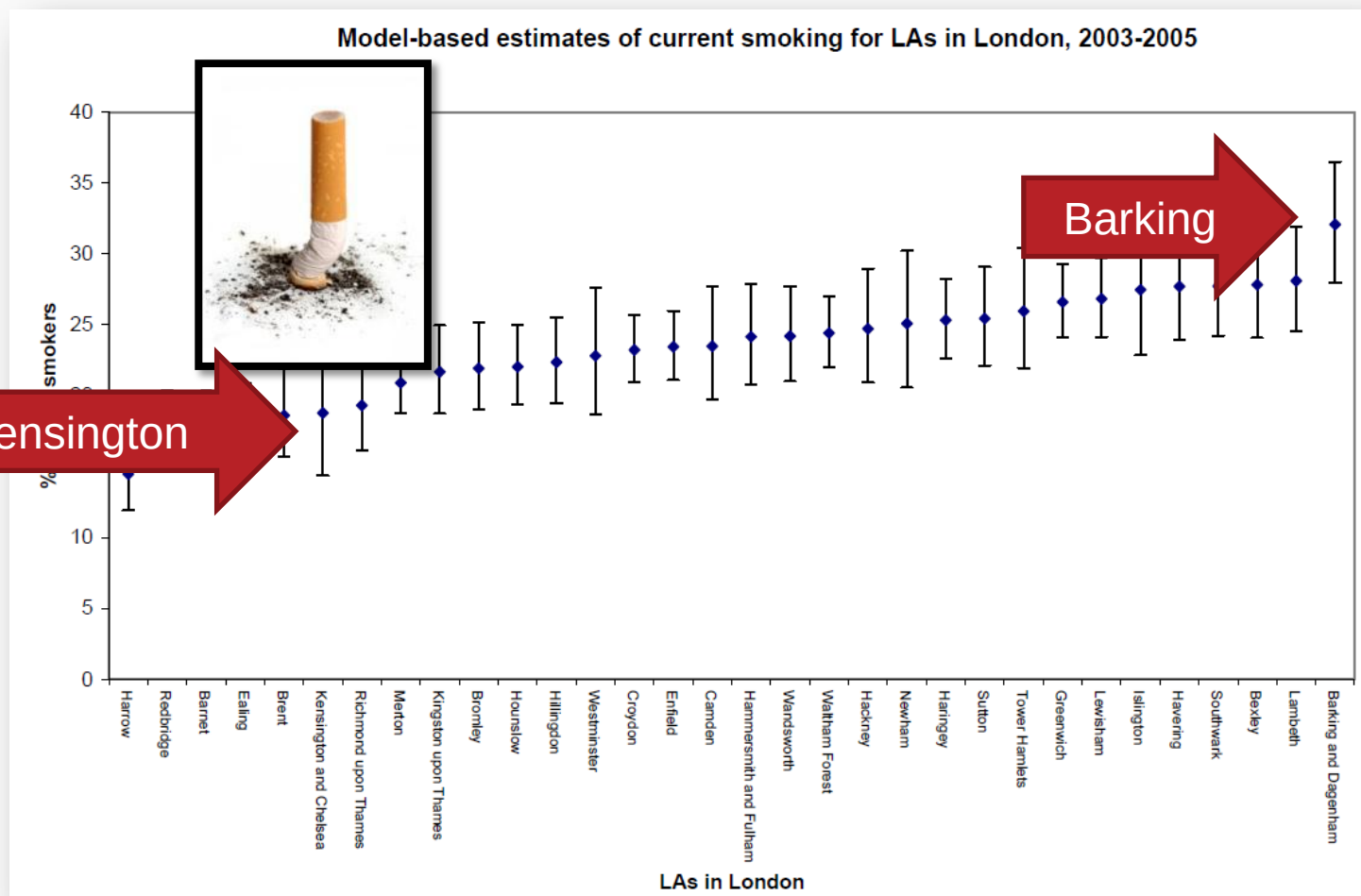


Source: <http://www.ic.nhs.uk/statistics-and-data-collections/population-and-geography/neighbourhood-statistics>



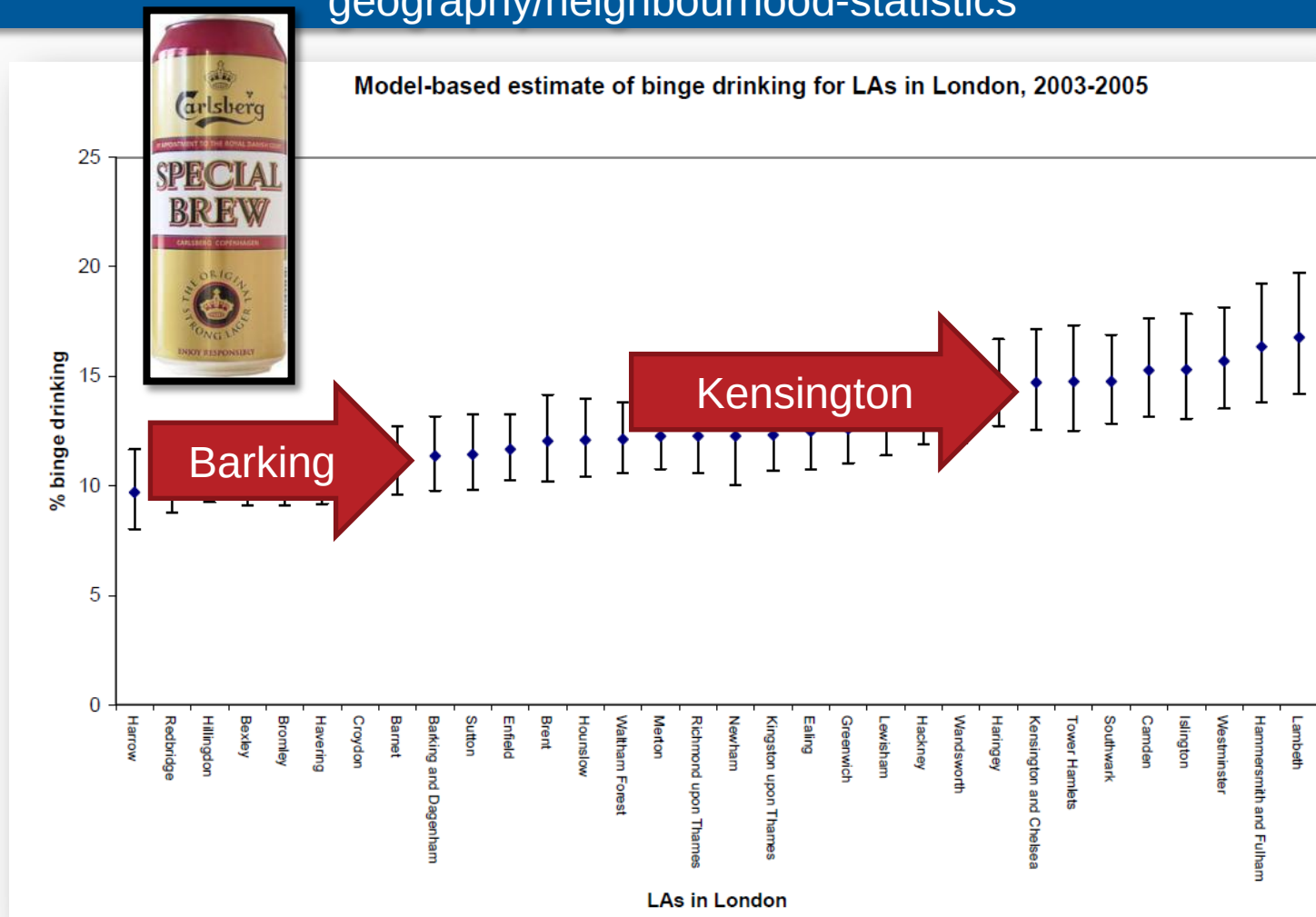


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Source: <http://www.ic.nhs.uk/statistics-and-data-collections/population-and-geography/neighbourhood-statistics>





Uncle Ian

Lives in some godforsaken
part of East London (Barking)



Uncle Mike

Lives in a nobby part of West
London (Kensington)

Protective clothing



Dot Cotton

Lives near Albert Square

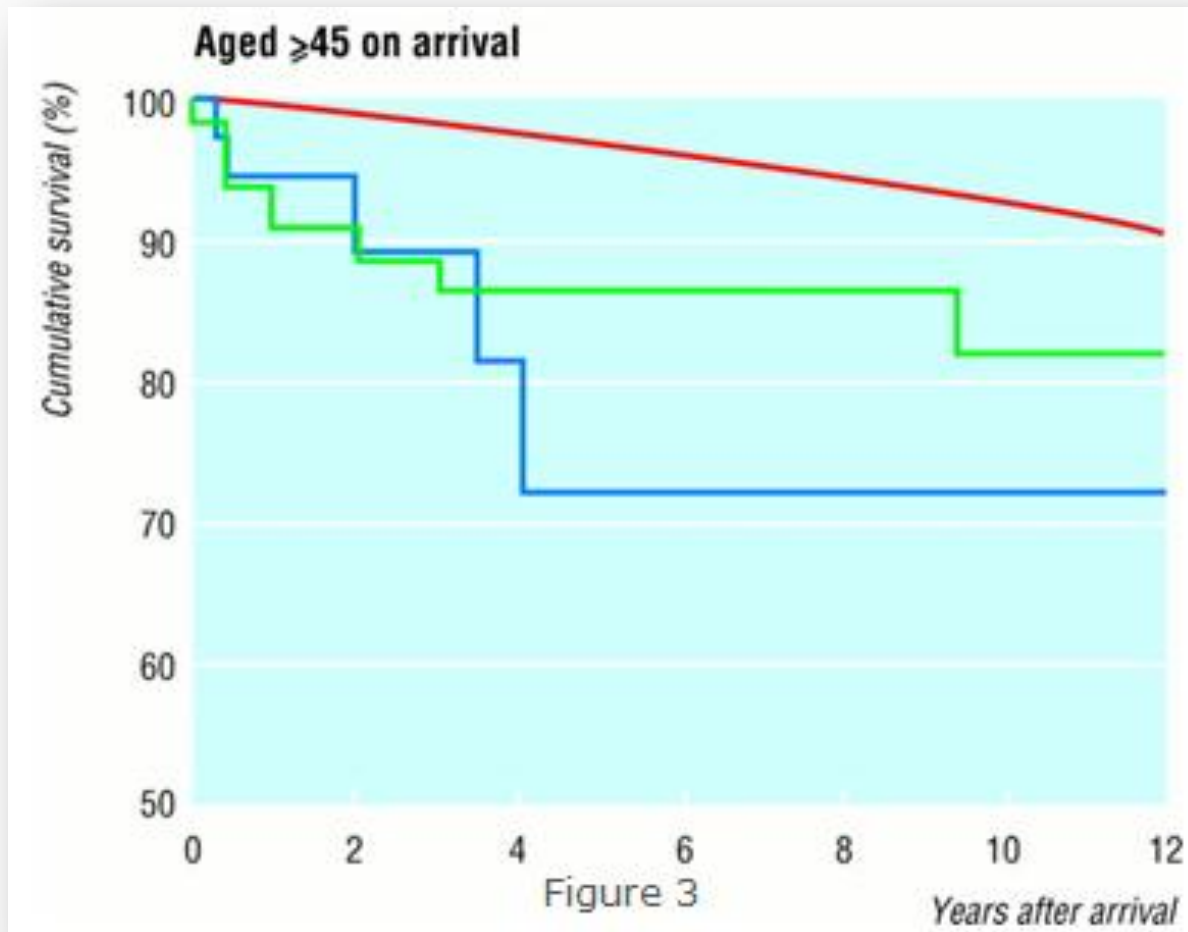


Ken Barlow

Lives on Coronation Street



Source: British Medical Journal



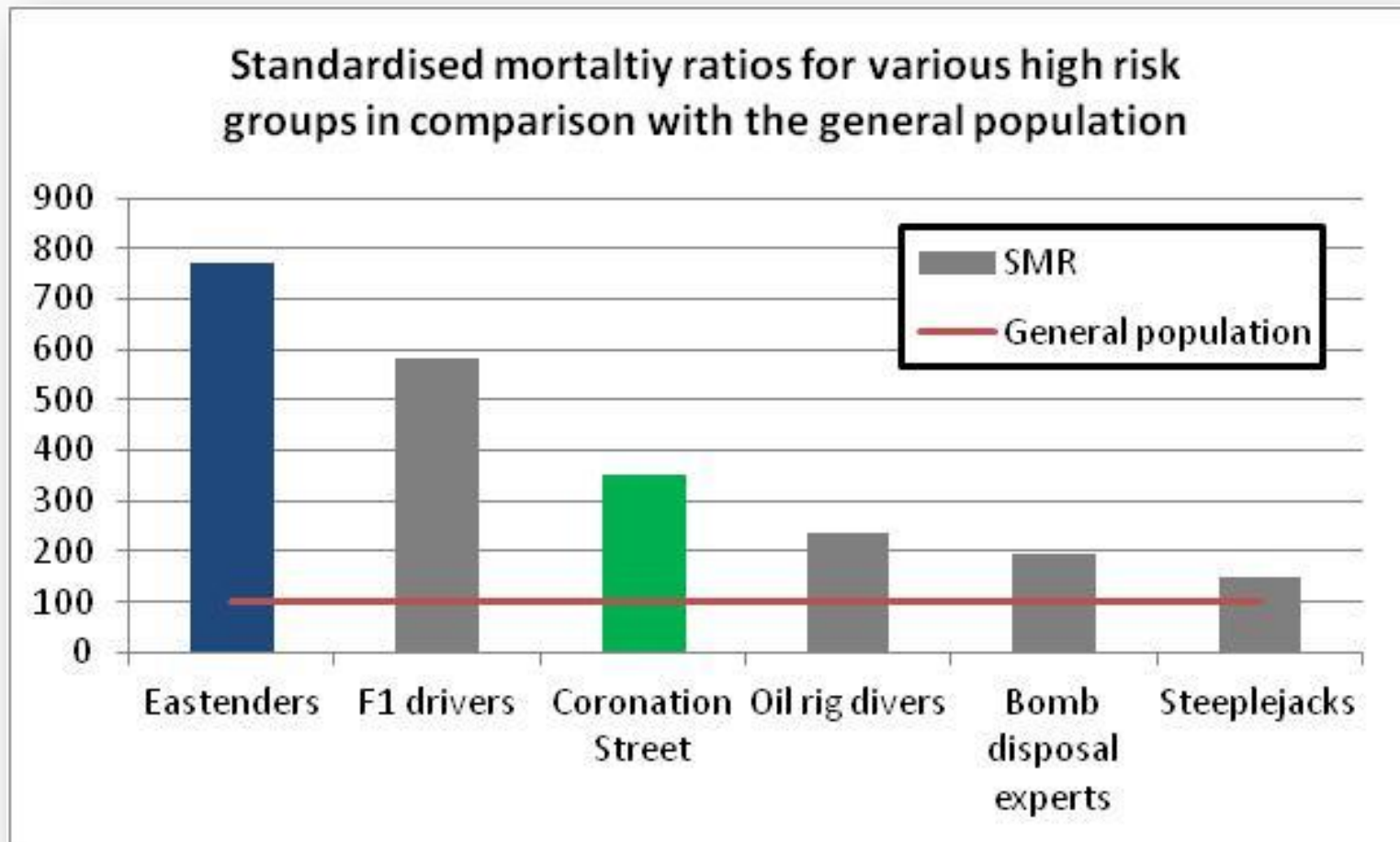
Population

Corrie

Benders



Source: British Medical Journal





Source: British Medical Journal

“Characters in these serials would be advised to wear **good protective clothing** (designed to withstand sharp implements, sudden impacts, and fire) and to receive regular counselling for the psychological impact of living in an environment **akin to a war zone.**”

Protective clothing



Dot Cotton

Lives near Albert Square



Ken Barlow

Lives on Coronation Street



Liverpool fan

Lives near Anfield



Man Utd fan

Lives near Old Trafford

Insurance application form disclosures



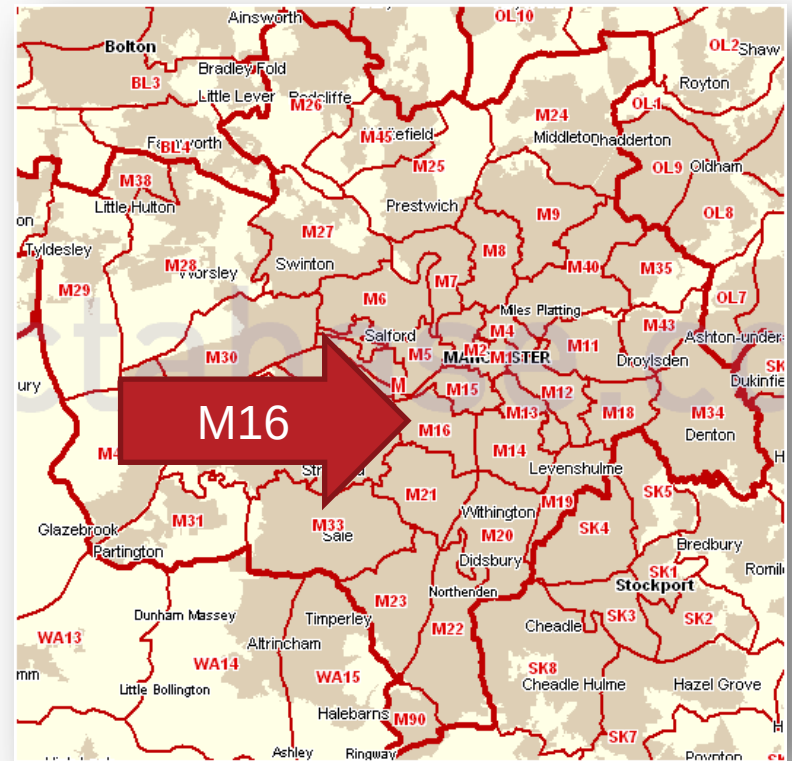
Source: Munich Re datawarehouse of insurance application forms

Component	Part	Example	Live codes ^[29]	Terminated codes ^[30]	Other codes	Total
postcode area	out code	YO	124	0	3	127
postcode district	out code	YO31	2,971	103	4	3,078
postcode sector	in code	YO31 1	10,631	1,071	4	11,706
postcode unit	in code	YO31 1EB	1,762,464 ^[30]	650,417	4	2,412,885
Postcode Addresses			approx. 27,000,000 ^[31]			

Anfield

Old Trafford

A detailed map of the L4 postcode area in Merseyside, UK. The map shows the coastline of the Wirral Peninsula to the west and the Mersey River to the east. Major roads are highlighted in red, including the A56, A50, and A5. Surrounding postcode areas are labeled, such as L37, L38, L29, L23, L22, L21, L30, L31, L32, L33, L34, L35, L36, L37, L38, L39, L40, L41, L42, L43, L44, L45, L46, L47, L48, L49, L50, L51, L52, L53, L54, L55, L56, L57, L58, L59, L60, L61, L62, L63, L64, L65, L66, L67, L68, L69, L70, L71, L72, L73, L74, L75, L76, L77, L78, L79, L80, L81, L82, L83, L84, L85, L86, L87, L88, L89, L90, L91, L92, L93, L94, L95, L96, L97, L98, L99, L00. The map also shows the locations of several towns and villages, including Southport, Ainsdale, Scarisbrick, Halsall, Bursough Industrial Estate, Newburgh, Ormskirk, Haskayne, Lydiate, Bickerstaffe, Skelmersdale, Formby, and Wallasey. The L4 area is highlighted in a darker shade of blue, and a large red arrow points to it from the left.

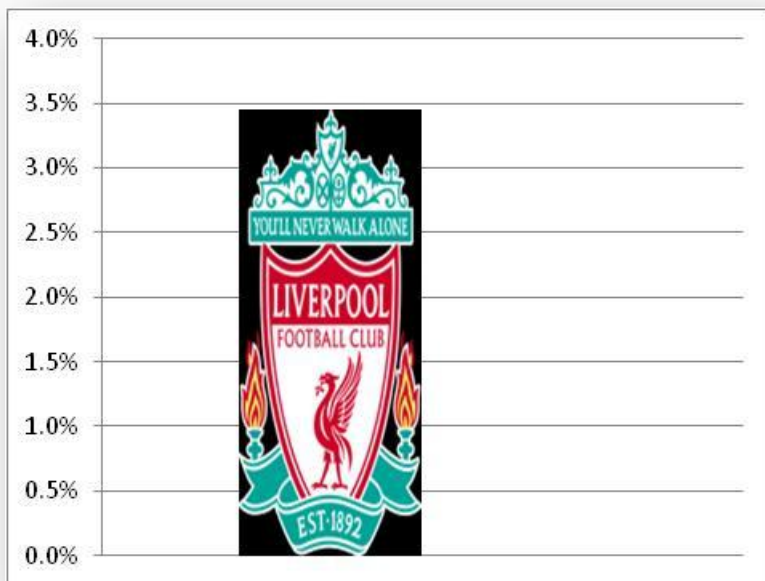


Insurance application form disclosures

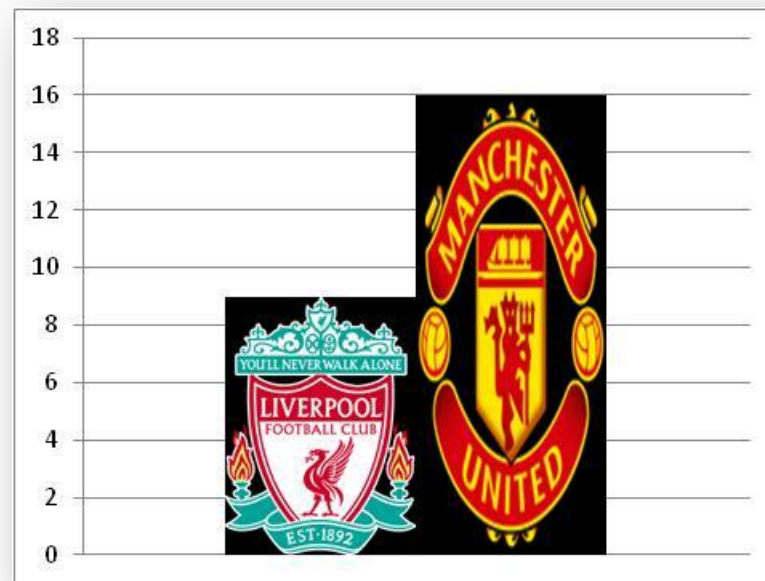


Source: Munich Re datawarehouse of insurance application forms

% of applicants disclosing
“depression”



Honours won since 2000





Source: Munich Re datawarehouse of insurance application forms



1871

Reading FC founded



1878

Man Utd FC founded



1892

Liverpool FC founded



Liverpool fan

Lives near Anfield



Man Utd fan

Lives near Old Trafford

100th birthday card



Old lady (Taurus)

Born in May



Old lady (Capricorn)

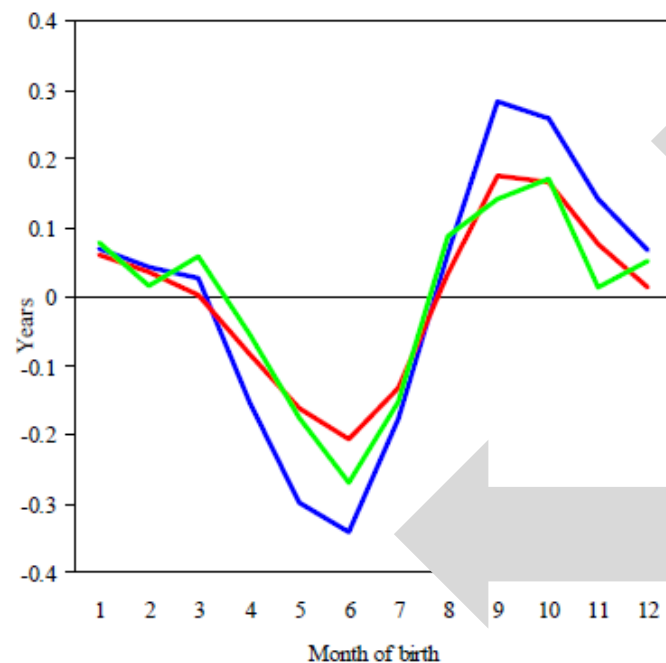
Born in December

What is the source of this?



Source: Max Planck Institute for Demographic Research (Doblhammer 2002)

Figure 2: Educational differentials in the deviations of mean age at death of decedents born in a specific month from average mean age at death; US death records 1989 to 1997, ages 50-100.



Age at death for those born in winter is higher...

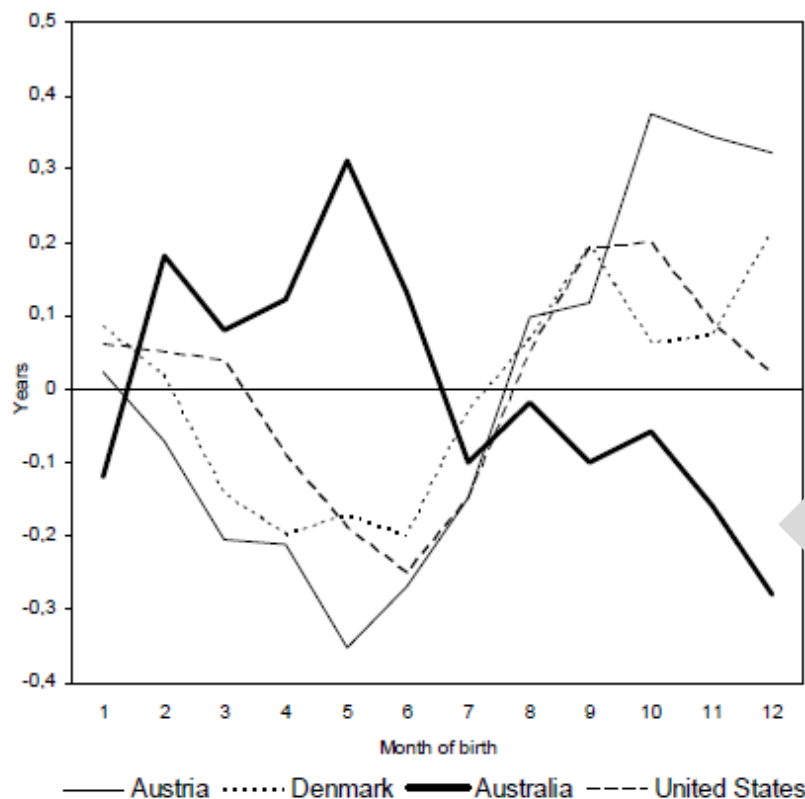
...than those born in the summer

What is the source of this?



Source: Max Planck Institute for Demographic Research (Doblhammer 2002)

Figure 1 Deviation from average remaining lifespan at age 50 for people born in a specific month in Austria, Australia, Denmark and the United States



The pattern for Australia is shifted by six months

100th birthday card



Old lady (Taurus)

Born in May



Old lady (Capricorn)

Born in December

Bunny suicides



Male chef

40 years old

Works as a chef



Female nurse

40 years old

Works as a nurse



Source: ONS Population trends



Teachers in higher education institutions

Chefs & cooks

Dentists
Veterinarians
Farmers

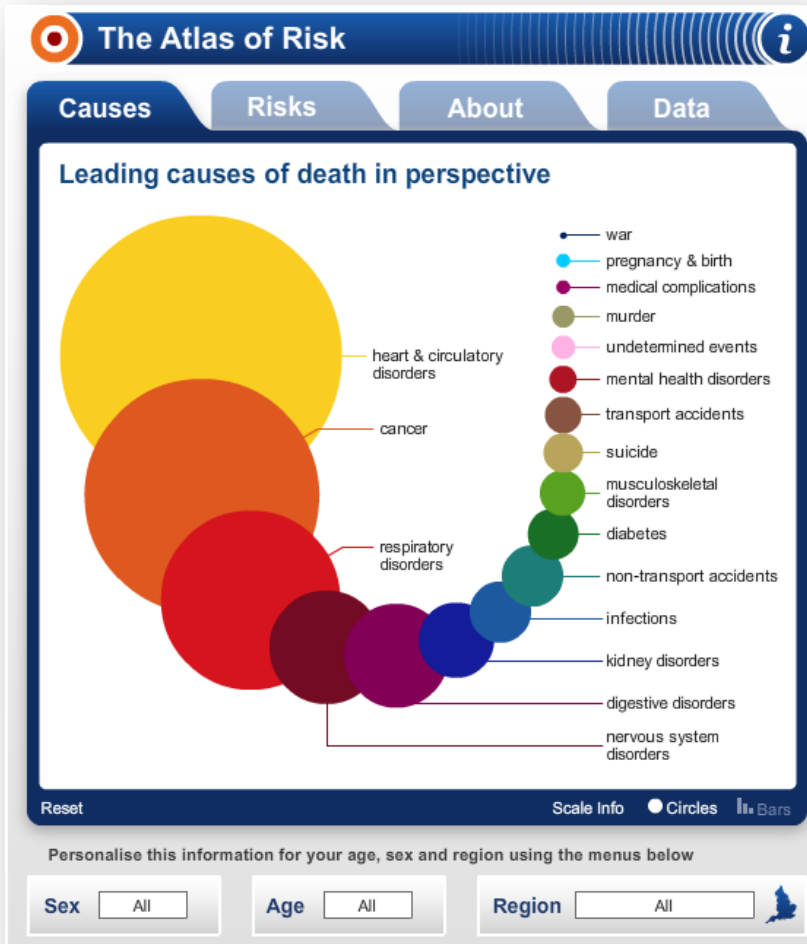


Chefs & cooks
Teachers in higher education institutions

Doctors
Veterinarians
Nurses



Source: <http://www.nhs.uk/Tools/Pages/NHSAAtlasofrisk.aspx>



Bubbles show relative frequency of different high level causes of death

You can filter results by age, sex and region



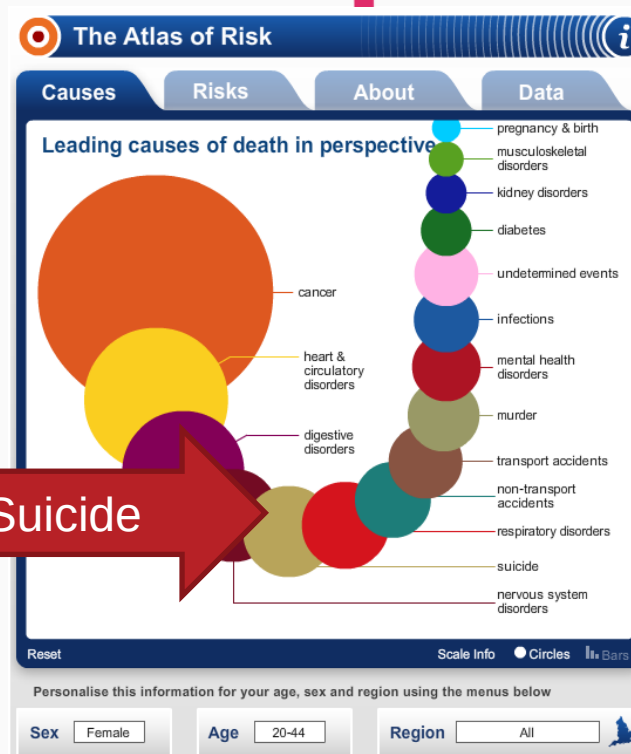
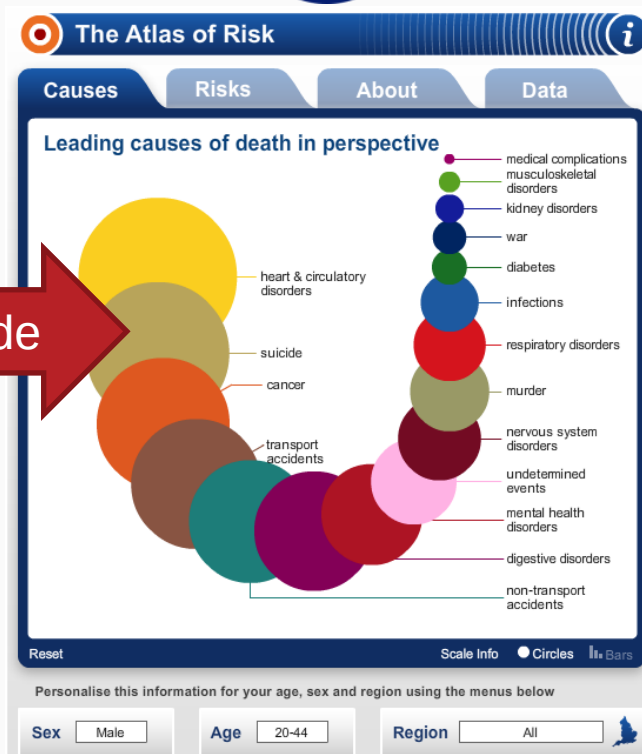
Source: <http://www.nhs.uk/Tools/Pages/NHSAAtlasofrisk.aspx>

Clicking on a bubble
drills down into sub-
causes





Source: <http://www.nhs.uk/Tools/Pages/NHSAAtlasofrisk.aspx>



Bunny suicides



Male chef

40 years old

Works as a chef



Female nurse

40 years old

Works as a nurse

Car safety sticker



West Country driver

Actuary

A spritely 31

Lives in ooh-arr land



Surrey driver

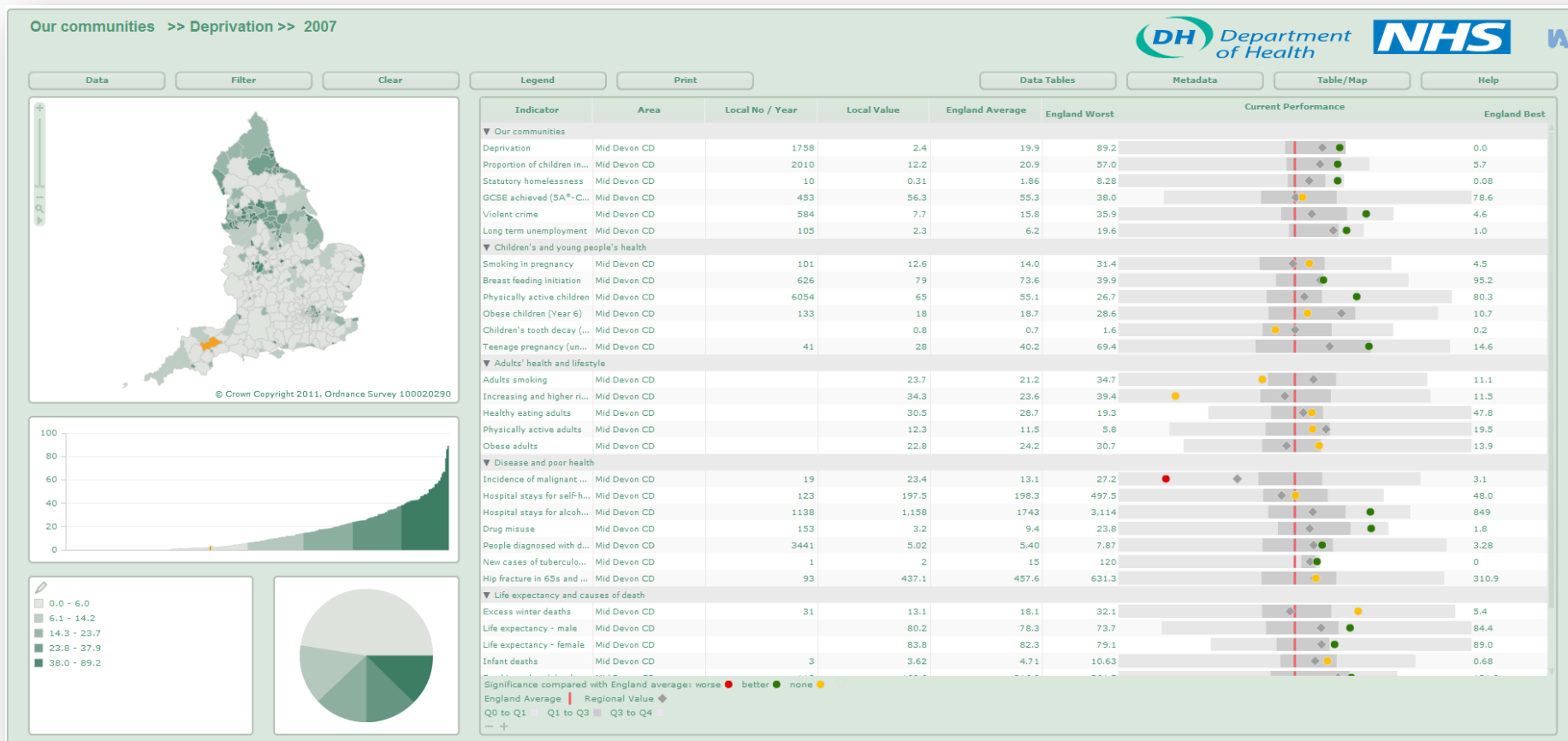
Actuary

A spritely 31

Lives in sophisticated Thurrey

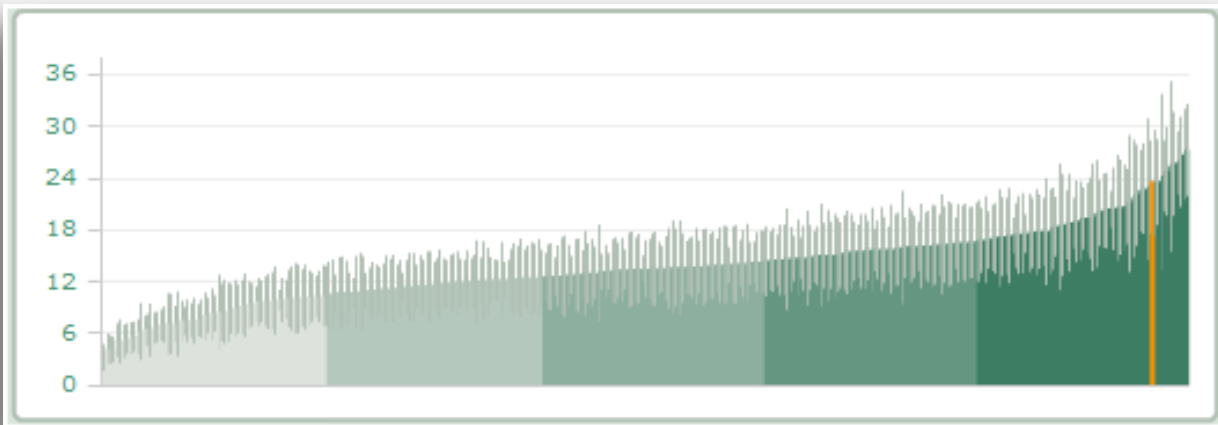


Source: http://www.apho.org.uk/default.aspx?QN=HP_INTERACTIVE2011



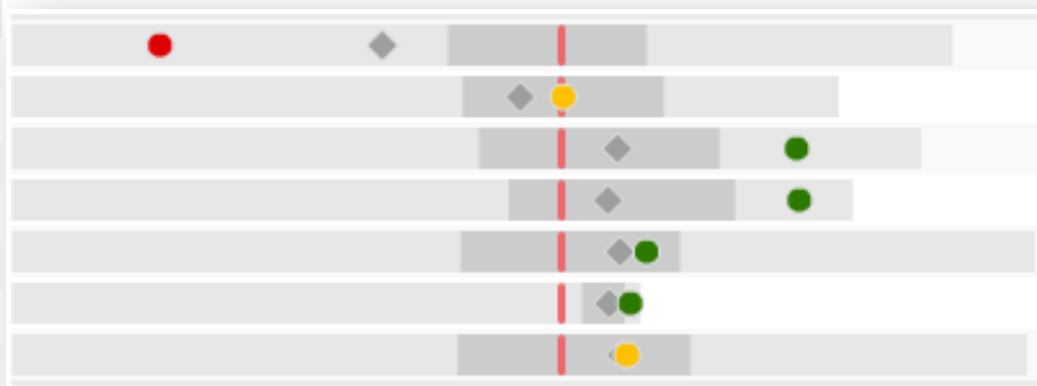


Source: http://www.apho.org.uk/default.aspx?QN=HP_INTERACTIVE2011



▼ Disease and poor health

Incidence of malignant ...	Mid Devon CD
Hospital stays for self-h...	Mid Devon CD
Hospital stays for alcoh...	Mid Devon CD
Drug misuse	Mid Devon CD
People diagnosed with d...	Mid Devon CD
New cases of tuberculo...	Mid Devon CD
Hip fracture in 65s and ...	Mid Devon CD





Source: http://www.apho.org.uk/default.aspx?QN=HP_INTERACTIVE2011



People killed or seriously injured
(per 100,000 resident population)

Elmbridge 39.4

Mid-Devon: 35.7

Car safety sticker



West Country driver

Actuary

A spritely 31

Lives in ooh-arr land



Surrey driver

Actuary

A spritely 31

Lives in sophisticated Thurrey



1 Never let an actuary choose your Christmas presents

2 Your actuarial colleagues know about lots of interesting data, some of which is very accessible

Journals

NHS Instant Atlas

Office of National Statistics

NHS Neighbourhood Stats

Insurance disclosures

NHS Atlas of Risk

3 Although some care is needed when using this data, it could be helpful in your day-to-day jobs



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